Introduction

The Bland-Altman plot (1986) compares two measurement methods by plotting their difference against their mean. It exposes systematic bias, proportional bias, and the 95 % limits of agreement within which most differences fall. It is the standard graphical summary for method-comparison studies and has largely replaced correlation-based summaries.

Prerequisites

Paired measurements; method-comparison basics.

Theory

For paired measurements (A_i, B_i): - **Bias**: mean difference $\bar{d} = \overline{A_i - B_i}$. - **Limits of agreement**: $\bar{d} \pm 1.96 \cdot s_d$.

If 95 % of differences lie within the limits and the limits are clinically acceptable, the two methods can be used interchangeably. Proportional bias shows as a trend in the scatter.

Assumptions

Differences are approximately normal; differences do not systematically depend on the mean (check with regression); replicates are handled appropriately if present.

R Implementation

```r
library(ggplot2)

set.seed(2026)
n <- 100
truth <- rnorm(n, 10, 2)
A <- truth + rnorm(n, 0, 0.5)            # method A
B <- truth + 0.3 + rnorm(n, 0, 0.5)      # method B (small bias)

df <- data.frame(A, B,
                 mean = (A + B) / 2,
                 diff = A - B)

bias <- mean(df$diff)
sd_d <- sd(df$diff)
loa  <- c(bias - 1.96 * sd_d, bias + 1.96 * sd_d)

ggplot(df, aes(mean, diff)) +
  geom_point(colour = "#2A9D8F") +
  geom_hline(yintercept = bias, linetype = 1) +
  geom_hline(yintercept = loa, linetype = 2) +
  labs(x = "Mean of A and B", y = "A - B",
       title = "Bland-Altman plot",
       subtitle = sprintf("Bias %.2f; 95%% LoA [%.2f, %.2f]",
                          bias, loa[1], loa[2])) +
  theme_minimal()
```

Output & Results

A scatter of differences vs means with solid bias line and dashed LoA; the simulated systematic bias of -0.3 is recovered.

Interpretation

"Bland-Altman analysis revealed a bias of -0.3 units with 95 % limits of agreement (-1.7, 1.1). If the clinically acceptable limit is +-2 units, the methods are interchangeable for most practical purposes."

Practical Tips

- Always show both bias and LoA; correlation alone does not reveal systematic bias.
- Check for proportional bias by regressing differences on means; a non-zero slope indicates non-constant bias.
- For replicated measurements per subject, adjust the LoA calculation (Bland-Altman 1999 extension).
- Report the clinical acceptance criterion **before** calculating LoA; otherwise post-hoc thresholding biases conclusions.
- Paired with ICC, Bland-Altman gives both a quantitative and visual summary of agreement.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale